

DRUMMOND ISLAND, MI, USA

WMO: 720375

Lat: 46.007N

Lon: 83.743W

Elev: 665

StdP: 14.35

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 54844

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.0	-4.5	-17.3	2.2	-9.7	-11.5	3.0	-3.2	18.7	24.4	17.0	25.1	2.5	340	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	82.1	68.6	79.7	67.2	77.1	65.7	71.6	78.4	69.9	75.4	68.3	74.0	6.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.8	112.1	75.1	68.0	105.5	73.2	66.1	98.7	71.2	35.9	78.4	34.3	75.3	32.9	73.4	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.6	14.3	12.4	-19.0	86.3	6.1	3.4	-23.3	88.8	-26.9	90.8	-30.3	92.7	-34.7	95.2		
			-19.4	74.3	6.0	1.9	-23.7	75.7	-27.2	76.8	-30.6	77.9	-34.9	79.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	17.0	17.5	26.4	37.8	51.2	60.4	66.3	65.2	58.6	46.9	35.7	24.7	
	DBStd	19.39	12.14	10.62	10.82	7.83	7.55	5.78	5.27	4.99	7.03	7.47	8.09	9.84	
	HDD50	4492	1024	911	734	370	74	3	0	0	9	153	431	783	
	HDD65	8436	1489	1331	1198	815	432	160	50	60	213	562	878	1248	
	CDD50	1735	0	0	1	5	112	314	504	472	267	57	3	0	
	CDD65	203	0	0	0	0	5	21	89	67	21	0	0	0	
	CDH74	1305	0	0	0	1	55	175	629	361	81	3	0	0	
	CDH80	201	0	0	0	0	5	25	120	43	8	0	0	0	
Wind	WSAvg	5.2	5.4	5.6	5.5	5.8	5.4	4.6	4.3	4.2	4.6	5.7	5.9	5.8	
Precipitation	PrecAvg	41.4	3.3	2.4	2.4	3.2	3.3	3.5	3.3	3.5	4.3	4.6	3.7	3.9	
	PrecMax	49.3	5.6	4.5	5.5	6.1	5.3	6.1	6.2	5.8	7.7	7.2	6.0	7.1	
	PrecMin	33.2	1.5	0.8	0.3	1.5	1.7	1.8	1.6	1.9	1.9	1.9	1.5	0.8	
	PrecStd	4.4	0.9	0.9	1.2	1.1	1.0	1.1	1.1	1.0	1.5	1.5	1.1	1.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	41.8	59.1	64.4	80.7	83.3	87.1	84.0	81.2	71.7	60.1	48.4	
		MCWB	41.2	39.1	47.6	50.8	65.3	68.2	70.5	71.2	69.3	62.4	53.8	47.4	
	2%	DB	37.5	36.5	49.7	59.2	74.7	79.0	83.1	81.0	76.3	65.8	53.8	42.5	
		MCWB	35.8	34.8	44.8	48.2	61.1	66.1	68.5	69.1	67.2	60.0	49.7	40.9	
	5%	DB	36.1	34.1	43.9	54.5	70.1	75.1	80.7	78.3	72.9	62.8	50.2	39.5	
		MCWB	34.6	32.7	38.9	44.7	58.6	63.0	67.5	67.0	64.9	57.3	46.9	37.5	
	10%	DB	33.7	32.2	39.4	50.3	65.0	72.2	77.4	75.3	70.3	59.2	46.6	37.1	
		MCWB	32.8	30.9	35.4	42.4	55.4	61.2	66.1	65.4	63.9	55.2	44.3	35.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.7	39.3	53.1	53.9	67.4	71.9	74.4	73.3	71.6	64.7	54.3	47.8
			MCDB	42.4	41.5	58.3	60.9	76.5	79.9	82.4	81.1	76.6	67.3	58.5	48.4
2%		WB	36.8	35.2	44.5	50.2	63.8	67.8	71.8	71.4	69.3	60.7	50.3	41.2	
		MCDB	37.7	36.3	48.1	56.8	71.4	75.2	78.8	78.1	73.7	65.0	53.1	42.1	
5%		WB	34.5	32.8	39.6	46.3	60.5	65.5	69.7	69.8	67.0	58.1	47.3	38.1	
		MCDB	35.5	33.5	43.6	52.3	67.2	72.6	76.1	74.4	71.4	61.7	49.5	39.2	
10%		WB	32.6	30.9	36.3	43.1	57.3	63.4	68.1	68.1	64.9	55.5	44.9	36.0	
		MCDB	33.4	31.8	38.9	48.8	63.6	69.4	74.8	72.9	69.1	59.0	46.6	37.1	
Mean Daily Temperature Range	5% DB	MDBR	14.3	15.3	16.5	15.9	19.8	19.0	18.7	17.1	16.7	13.5	10.8	11.4	
		MCDBR	9.8	13.2	18.5	24.5	26.4	23.5	22.1	20.1	18.8	18.3	16.0	11.3	
		MCWBR	9.4	11.7	13.1	14.5	14.9	11.9	10.4	9.8	10.6	12.6	12.4	10.6	
	5% WB	MCDBR	9.4	13.0	17.5	21.7	23.2	19.4	18.6	16.7	16.1	16.1	14.7	11.0	
		MCWBR	9.2	12.3	13.0	14.7	14.3	11.6	10.3	9.7	10.9	12.1	12.2	10.7	
Clear-Sky Solar Irradiance	taub	0.290	0.293	0.319	0.354	0.374	0.395	0.394	0.384	0.366	0.331	0.304	0.285		
	taud	2.383	2.365	2.370	2.334	2.374	2.349	2.372	2.401	2.439	2.503	2.517	2.457		
	Ebn at Noon	258	281	289	286	283	277	275	274	270	265	253	249		
	Edh at Noon	23	28	33	37	37	38	37	35	31	25	20	19		
All-Sky Solar Radiation	RadAvg	375	663	1125	1458	1767	1906	1907	1614	1240	733	394	281		
	RadStd	43	73	108	145	160	148	89	97	108	80	51	28		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page